

A machine learning approach for estimating intrinsic capacity age and its associations with multimorbidity and geroprotective agents

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Abstract

Background and Objectives: Aging is associated with functional decline and multimorbidity, highlighting the need for holistic biomarkers to monitor healthy aging. The aim of this study was to validate intrinsic capacity age (IC-age) as a biomarker of aging and to examine its association with multimorbidity and geroprotective agents.

Research Design and Methods: A cross-sectional study was conducted with data from 48,068 participants aged ≥60 years from the 9th wave of the Survey of Health, Ageing and Retirement in Europe (2021–2022). Random forest regression was used to train a model predicting IC-age based on biomarkers (cognitive, psychological, sensory, vitality, locomotion) and demographic factors.

Results: IC-age showed a prediction error of 5.3 years ($r = 0.55$). Biomarkers for vitality (handgrip strength), cognitive (verbal fluency and memory), and sensory (hearing aid use) domains were important contributors. General linear models assessed associations with multimorbidity, physical activity, and smoking. Intrinsic capacity age was significantly higher in individuals with multimorbidity and smokers compared with healthy individuals. Physical activity exhibited a protective effect on IC-age, with vigorous activity showing a particularly pronounced benefit in women.

Discussion and Implications: This model demonstrates that IC domains can estimate biological age and distinguish individuals based on their comorbidities. It also underscores the role of physical activity as a key geroprotective factor, with vigorous physical activity in females with comorbidities showing the most pronounced protective effect on IC-age. These results validate the concept of IC-age as a comprehensive measure of aging and highlight its potential to inform personalized interventions and public health strategies.

Keywords: Biomarkers, Aging, Machine learning, Physical fitness, Healthy lifestyle

The world's population of older adults is expected to grow substantially, presenting significant global challenges (Beard et al., 2016; World Health Organization, 2015). Aging, as a process of accumulating life consequences such as molecular and cellular damage, leads to functional decline, chronic diseases, and ultimately mortality (Moqri et al., 2023). This creates a pressing need for many countries to promote healthy aging and address its health, social, and economic impacts. Aging, as a progressive process, is characterized by declining vitality, sensory capacity, cognitive function, and locomotion, which together constitute an individual's "intrinsic capacity" (Bautmans et al., 2022). Intrinsic capacity (IC) is defined by the World Health Organization (WHO) as "the composite of all

the physical and mental capacities that an individual can draw on" at any point in time (World Health Organization, 2015). The WHO developed this term as a holistic metric to assess and monitor overall functional health in aging populations. By emphasizing IC, the WHO shifts the focus from disease-centered models to a broader understanding of health, encouraging targeted interventions to promote healthy aging (Bautmans et al., 2022; Cesari et al., 2018), defined as "the process of developing and maintaining the functional ability that enables wellbeing in older age" (Cesari et al., 2018). Advances in aging research have highlighted the role of geroprotectors—including lifestyle factors (e.g., physical activity, sleep health, and healthy eating) and pharmacological interventions (e.g., metformin)—in

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maintaining homeostasis, delaying functional decline, and promoting longer, healthier lives (Beyene et al., 2024; Moskalev et al., 2017; Trendelenburg et al., 2019). Among geroprotectors, exercise is the most accessible and cost-effective in impacting healthy aging (Elliehausen et al., 2023).

Given the importance of interventions to promote healthy aging, studies have focused on determining biological age, defined by age-related molecular and cellular changes, as a metric for developing health policies for longevity (Diebel & Rockwood, 2021; Moqri et al., 2023, 2024). In contrast, chronological age refers to the time since birth. This distinction provides valuable insights into functional decline and temporal progression throughout life (Moqri et al., 2023). Biomarkers of aging for predicting biological age has primarily focused on genetic markers, such as telomere length (Diebel & Rockwood, 2021; Galkin et al., 2020; Moqri et al., 2024). However, functional performance markers could offer a more practical and accessible means of assessing biological aging (Bohannon, 2019; López-Bueno et al., 2022; McGrath et al., 2018). Modeling approaches, such as Random Forest algorithms, have shown promise in estimating biological age, achieving margins of error of 3–6 years compared to chronological age based on diverse biomarkers, including blood profiles and genetic data (R. Chen et al., 2023; Galkin et al., 2020; Mamoshina et al., 2018). Despite their accuracy, their complexity and cost limit feasibility for routine clinical use. Thus, there is an urgent need for cost-effective, holistic methods to estimate biological age.

Considering the importance of determining a holistic aging metric, this study proposes an integrative model to estimate “intrinsic capacity age (IC-age)” based on a composite metric that consolidates the five functional domains of IC (i.e., cognition, locomotion, vitality, sensory, and psychological) (World Health Organization, 2015) into a single, novel biomarker. IC-age may have the potential to serve as a comprehensive measure of aging, offering insights into personalized interventions and public health strategies. Unlike traditional biological age, which is typically estimated using molecular or cellular biomarkers, IC-age is proposed as a clinically relevant functional biomarker of aging. The aim of this study is to validate IC-age as a biomarker of aging by comparing its prediction error between healthy individuals and those with multimorbidity. Additionally, we examine the association between IC-age and specific chronic conditions, as well as the potential protective influence of geroprotective factors such as physical activity. We hypothesize that IC-age will differ between healthy individuals and subjects with multimorbidity, with geroprotector agents influencing this differentiation. This holistic approach could advance our understanding of IC biomarkers in aging, inform global policies, and support interventions to promote healthy aging.

Method

Design and participants

A cross-sectional analysis was conducted using data from the 9th wave of the Survey of Health, Ageing and Retirement in Europe (SHARE), a representative survey of individuals aged 50 years and older in 27 European countries. The 9th regular wave of SHARE was conducted between October 2021 and September 2022 (Börsch-Supan et al., 2013). SHARE uses a multistage stratified sampling design (Wave 9, n.d.). Participating countries are divided into different strata based on their

geographical area, and municipalities or postal codes within these strata serve as the primary sampling units. Data are collected every 2 years through computer-assisted, face-to-face, in-home interviews using ex ante harmonized questionnaires. New respondents are added to each wave to compensate for attrition. The target population for SHARE is all individuals aged 50 and over at the time of sample collection who are permanent residents of the respective SHARE country. Individuals who are incarcerated, hospitalized, out of the country, do not speak the language(s) of the country, or have moved to an unknown address during the entire survey period are excluded from the baseline or follow-up samples. Partners living in the same household as the target population are interviewed, regardless of age. Written informed consent was obtained from all those willing to participate. Ethical approval for the collection of the data used in this study was granted by the SHARE project. This study was reported according to the Strengthening the Reporting of Observational Studies in Epidemiology statement (von Elm et al., 2014). This dataset was selected because it includes variables related to cognition, locomotion, vitality, sensory, and psychological factors, which may potentially represent IC-age, see below.

Biomarkers employed in the development of the IC-age model

Data were collected for different biomarkers reflecting the five domains of IC (cognition, locomotion, vitality, sensory, and psychological) proposed by previous studies (Beard et al., 2016, 2022). These domains were used to develop a predictive model of IC-age (see below), defined as the estimated age derived from functional biomarkers that best approximate chronological aging. Based on these domains and the available variables in the 9th regular wave of SHARE, four variables were selected for cognitive capacity, two for psychological capacity, three for sensory capacity, two for vitality, and one for locomotor capacity. This uneven distribution resulted from the varying availability of variables across the five domains in the dataset. (1) For the *cognitive domain*, four cognitive measures were used: verbal fluency, immediate recall, delayed recall, and calculation (i.e., arithmetic), based on the Modified Telephone Interview for Cognitive Status (Knopman et al., 2010). For verbal fluency, respondents were asked to name as many words as possible from the semantic category “animals” in 60 s. The final score was the total number of correctly named words. Immediate and delayed recall was assessed with the 10-word recall test, which consists of listening to a list of ten words and being asked to retrieve the list immediately after encoding (immediate recall) and another list after a delay (delayed recall). The different lists were randomly assigned to respondents. The total score for these two tests ranged from 0 to 10 points, corresponding to the number of words the respondent was able to recall correctly. Finally, numeracy was assessed as the participant’s mathematical performance on a subtraction task. The participant was first asked, “One hundred minus seven equals what?” and then asked to subtract seven from the previous result four more times. The resulting score, which represented the number of correct answers, ranged from 0 to 5; (2) The *psychological domain* was assessed using the depression subscale of the EURO-D instrument: “In the last month, have you been sad or depressed?” (Response: yes/no) (Prince et al., 1999). The total EURO-D score was also included as a biomarker; (3) The *vitality domain* included measurement of

handgrip strength and the difference between the maximum values of the left and right hands, using a hand-held dynamometer (Smedley, S Dynamometer, TTM, Tokyo, 0–100 kg). Each hand was measured twice under the guidance of trained interviewers. Participants held the upper arm perpendicular to the trunk, with the elbow in 90° flexion and the wrist in neutral position while standing or sitting. Trained interviewers provided standardized instructions to press the dynamometer with maximal effort for 2 s. The handgrip strength value was defined as the maximum value of either hand for the present analyses; (4) For the *sensory domain*, participants self-reported their level of eyesight distance and eyesight reading using a Likert-type scale, with response options: (1) excellent, (2) very good, (3) good, (4) fair, and (5) poor. Also, participants were categorized according to whether they had a hearing aid or not; (5) In the *locomotion domain*, participants reported having or not having difficulty walking across a room. Participants who selected the options “refused” or “don’t know” were excluded from this study. Sex was included as an adjustment variable in the model due to its established association with IC. Evidence from a large multicounty cohort using the 5th regular wave of SHARE data revealed significant sex differences in IC and its subdomains, with men showing higher scores than women in all domains except cognition (M. Chen et al., 2025). Finally, although Geographic Region is not part of the formal IC domains, it was included in the model as an adjustment variable to account for regional disparities in healthcare access, lifestyle factors, and social determinants that may indirectly influence IC (M. Chen et al., 2025). This aligns with the conceptual framework of IC, which also considers the combination of relevant environmental characteristics and their interaction with the individual (Beard et al., 2016).

Model training of IC-age and optimization

To predict the IC-age, we trained a Random Forest regression model using the MATLAB software 2023b using the *fitensemble* function. The dataset was filtered to include individuals aged 60–99 years, excluding those with incomplete or missing responses, as well as those with handgrip strength values

recorded as 0. Additionally, individuals from a specific country (e.g., Israel) were excluded. The total sample analyzed included 48,068 participants. Predictor variables included all intrinsic biomarkers and demographic factors (e.g., sex, geographic region), while the response variable was chronological age. The dataset was randomly divided into training (80%) and testing (20%) sets using a hold-out cross-validation strategy with a fixed random seed for reproducibility.

We performed Bayesian Optimization over 30 iterations to tune the following hyperparameters: (1) the minimum number of samples per leaf node, evaluated within the range [5, 100], (2) the maximum number of splits per decision tree was evaluated within the range [20, 200], (3) the total number of trees in the forest evaluated within the range [100, 2000]. The final objective function minimized the training set’s Mean Absolute Error (MAE) across 10-fold cross-validation. The optimal hyperparameters identified for the Random Forest model were a minimum leaf size of 100, a maximum number of splits of 200, and 1925 trees. A broader range of hyperparameters (i.e., [5, 200] for leaf size, [20, 300] for splits, and [100, 3000] for the number of trees) was also explored in subsequent analyses, showing no significant improvement in error reduction (0.2%), confirming the stability of the initially selected parameters. The final model incorporates the optimal hyperparameters, and its performance was evaluated on the test set, reporting metrics such as MAE and Pearson correlation between the predicted IC-age and actual chronological age. The final model demonstrated a moderate positive association between chronological age and predicted IC-age, with a correlation coefficient of $r=0.55$ and a MAE of 5.3 years (Figure 1). This accuracy is comparable to other studies on biological age prediction, which report MAE values ranging between 3.5 and 7.7 years across various modalities such as brain age, retinal age, and blood-based biomarkers (R. Chen et al., 2023). The top five contributing variables were handgrip strength, verbal fluency, memory first, geographic region, and have hearing aid, highlighting the importance of vitality, cognitive, and sensory health as predictors of IC-age, as well as the interaction between individuals and their environments.

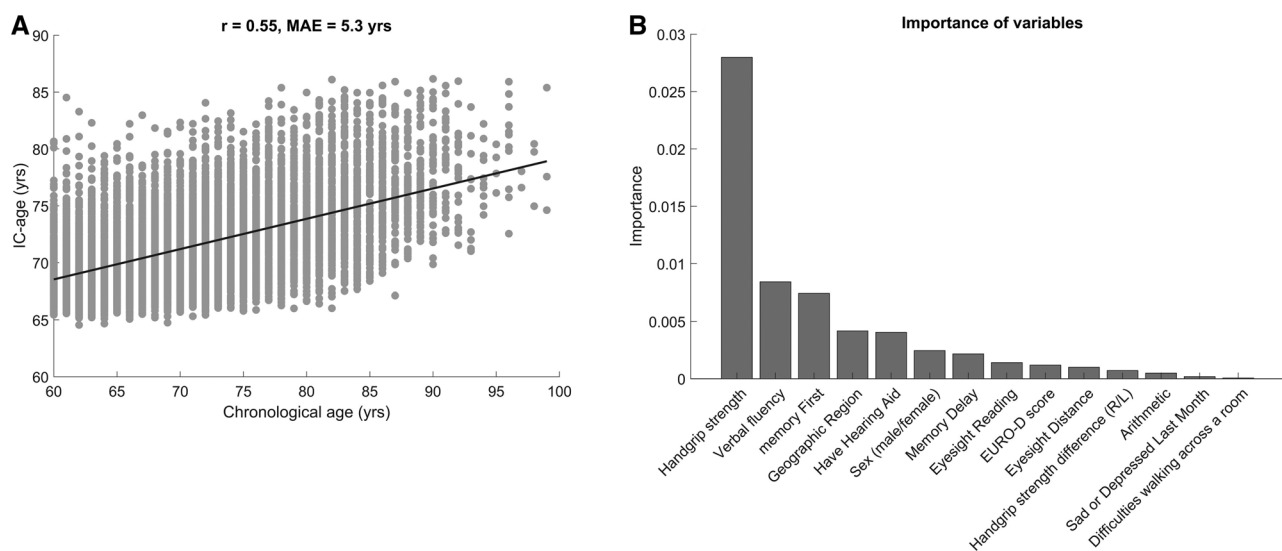


Figure 1. (A) Relationship between chronological age and estimated IC-age. (B) The importance variables of model prediction of IC-age.

Note. Bars represent the predictor variable importance of the model. MAE = mean absolute error; R = right; L = left; EURO-D = European depression scale; IC-age = intrinsic capacity age.

The final Random Forest model was also compared to other tree-based algorithms, specifically Decision Trees and Boosted Trees, using 10-fold cross-validation. We evaluated predictive performance based on MAE and Pearson correlation. To test statistical differences, we employed repeated-measures ANOVA with post hoc Bonferroni pairwise comparisons across folds. Random Forest showed superior accuracy and robustness compared to the other models (see [online supplementary material Figure S1](#)).

Chronic diseases

Past or current chronic diseases were self-reported by participants using the following question: “Has a doctor ever told you that you had/have any of the diseases on this card? This means that a doctor has told you that you have this condition and that you are either currently being treated for it or bothered by it.” The card in question had the following options “Heart attack, including myocardial infarction or coronary thrombosis, or any other heart problem, including congestive heart failure,” “High blood pressure or hypertension,” “High blood cholesterol,” “Stroke or cerebrovascular disease,” “Diabetes or high blood sugar,” “Chronic lung disease, such as chronic bronchitis or emphysema,” “Cancer or malignant tumor, including leukemia or lymphoma, Parkinson’s disease,” “Hip fracture,” “Alzheimer’s disease, dementia, organic brain syndrome, senility or other serious memory impairment,” “Rheumatoid arthritis,” “Osteoarthritis or other rheumatism,” and “Other condition not specified.” Participants who selected the options “refused” or “don’t know” were excluded from this study.

Statistical analysis

The resulting model was retained for subsequent statistical analyses across different age subgroups, comorbidity categories, and levels of physical activity, as well as other geroprotective factors such as sleep quality and dietary habits (e.g., fruit and vegetable consumption), and unhealthier lifestyle behaviors such as smoking. Comorbidities were divided into five categories: healthy, one disease, two diseases, three diseases, and more than four diseases. Physical activity frequency was categorized separately for vigorous and moderate activities into four groups (i.e., more than once a week, once a week, one to three times a month, and hardly ever or never). Sleep trouble was categorized as Yes/No, frequency of fruit and vegetable intake as Every day, 3–6 times a week, twice a week, once a week, or less than once a week, and smoking status as Yes/No.

A general linear model was applied to analyze the IC-age, specifically: (1) The difference between healthy individuals and subjects with varying numbers of diseases, based on four age categories (60–69; 70–79; 80–89; +90), adjusted for sex and geographic region (as defined by the United Nations). (2) The relationship between geroprotectors and the number of diseases, adjusted for sex and geographic region. A post hoc analysis with Bonferroni correction was performed for multiple comparisons. Data were reported as mean and 95% CI. A p -value threshold of .05 was considered statistically significant.

Results

A total of 48,068 participants were included. Participants had a mean age of 71.6 ($SD=7.7$) years and 55.6% were female. The characterized data are shown in [Table 1](#).

Comorbidities and IC-age

The number of diseases significantly influenced the IC-age. The general linear model adjusted by geographic region revealed a significant main effect of age group [$F_{(3, 159)} = 3034.1, p < .001$], indicating substantial differences in the number of diseases across age categories. Additionally, the number of diseases showed a significant main effect [$F_{(4, 159)} = 51.6, p < .001$]. A significant interaction between the age group and the number of diseases was also observed [$F_{(12, 159)} = 2.09, p < .001$]. Sex did not reach statistical interaction with factors [$F_{(12, 159)} = 1.65, p = .071$]. The post hoc comparison revealed significant differences between healthy individuals and those with one or more comorbidities across all age groups up to 90 years old, where the differences were no longer consistently significant ([Figure 2](#)). For p -values of post hoc comparison, see [online supplementary material Table S1](#).

Geroprotectors, lifestyles, and IC-age

Regarding the impact of geroprotectors, significant effects were observed for the frequency of vigorous physical activities [$F_{(3, 159)} = 638.5, p < .001$], along with a significant interaction between the number of comorbidities and the frequency of vigorous physical activities [$F_{(12, 159)} = 2.53, p = .002$]. Additionally, a significant three-way interaction was found between sex with the frequency of vigorous physical activities and the number of comorbidities [$F_{(12, 159)} = 1.91, p = .029$]. The post hoc analysis revealed that the frequency of vigorous physical activities had a protective effect, reflected in a lower IC-age, particularly in females ([Figure 3](#)). For three-way interaction visualization and p -values of post hoc comparison, see [online supplementary material Figure S2](#) and [Tables S2](#) and [S3](#). Notably, among individuals with more than four comorbidities, males showed no significant differences in IC-age between those who engaged in vigorous physical activity hardly ever and those who did so more than once a week, whereas females exhibited such differences. These findings suggest that females may be more sensitive to the impact of not engaging in vigorous physical activity on IC-age in the presence of comorbidities.

Regarding moderate physical activity significant effects were observed for the frequency of physical activities [$F_{(3, 159)} = 861.0, p < .001$], along with a significant interaction between the number of comorbidities and the frequency of physical activities [$F_{(12, 159)} = 2.03, p = .018$]. Non-significant three-way interaction was found between sex with the frequency of moderate physical activities and the number of comorbidities [$F_{(12, 159)} = 1.52, p = .107$]. Post hoc analyses revealed that individuals engaging in moderate physical activity more than once a week had significantly lower IC-age across all comorbidity levels ([Figure 4](#)). For p -values of post hoc comparison, see [online supplementary material Table S4](#).

Smoking habits have a negative impact on the IC-age among the number of comorbidities. Significant effects were observed for the number of comorbidities [$F_{(4, 79)} = 614.5, p < .001$] and smoking status [$F_{(1, 79)} = 338.3, p < .001$]. Additionally, a significant interaction between smoking status and the number of comorbidities was found [$F_{(4, 79)} = 11.7, p < .001$]. A non-significant three-way interaction was observed between sex, smoking status, and the number of comorbidities [$F_{(4, 79)} = 2.4, p = .050$]. The post hoc analysis revealed that smoking was associated with a significantly negative effect of age compared to non-smoking across all levels of comorbidities ([Figure 5](#)). For p -values of post hoc comparison, see [online supplementary material Table S5](#).

Table 1. Characteristics of participants (*n* = 48,068).

Characteristics	<i>n</i> (%)	Mean (<i>SD</i>)
Age (y)		71.6 (7.7)
Sex		
Male	21,344 (44.4%)	
Female	26,724 (55.6%)	
Geographic Region		
Eastern Europe	9,271 (19.3%)	
Northern Europe	10,212 (21.2%)	
Southern Europe	14,217 (29.6%)	
Western Europe	14,368 (29.9%)	
Ever smoked daily		
No	28,480 (59.2%)	
Yes	19,588 (40.8%)	
Fruit and vegetable consumption		
Every day	36,133 (75.2%)	
3–6 times a week	8,981 (18.7%)	
Twice a week	1,970 (4.1%)	
Once a week	593 (1.2%)	
Less than once a week	391 (0.8%)	
Moderate physical activity		
Hardly ever, or never	5,695 (11.8%)	
One to three times a month	3,378 (7.0%)	
Once a week	6,536 (13.6%)	
More than once a week	32,459 (67.5%)	
Vigorous physical activity		
Hardly ever, or never	21,126 (44.0%)	
One to three times a month	5,507 (11.5%)	
Once a week	6,809 (14.2%)	
More than once a week	14,626 (30.4%)	
Sleep problems		
Yes	16,487 (34.3%)	
No	31,581 (65.7%)	
Number of diseases		
0	13,760 (29.0%)	
1	16,757 (35.3%)	
2	10,911 (23.0%)	
3	4,544 (9.6%)	
4+	1,555 (3.3%)	
Intrinsic capacity biomarkers		
Cognitive capacity: Verbal fluency (number of words)		20.6 (7.7)
Cognitive capacity: Immediate recall (0-10)		5.3 (1.7)
Cognitive capacity: Delayed recall (0-10)		3.8 (2.1)
Cognitive capacity: Numeracy (0-5)		4.2 (1.4)
Psychological capacity: EURO-D Score (0-12)		2.33 (2.2)
Sensory capacity: Eyesight distance (1-5)		2.57 (1.0)
Sensory capacity: Eyesight reading (1-5)		2.64 (1.0)
Sensory capacity: Hearing aid (yes)	4,636 (9.6%)	
Vitality: Handgrip strength (kg)		31.6 (11.0)
Vitality: Handgrip strength asymmetry (kg)		3.9 (3.3)
Locomotion: Difficulty walking in a room (yes)	681 (1.4%)	

Note. *n* = number; *SD* = standard deviation; *y* = years; kg = kilograms; EURO-D = European depression scale.

Notably, no significant interaction was found between the number of diseases and sleep problems [$F_{(4,159)} = 1.67, p = .153$], or fruit and vegetable consumption [$F_{(16,159)} = 1.45, p = .110$].

Overall, these findings highlight the positive role of physical activity as a geroprotective agent in preserving the IC-age. In contrast, smoking habits were associated with a negative impact on IC-age, independent of sex.

Discussion

This study aimed to validate the IC-age as a novel biomarker of aging and examine its association with multimorbidity and geroprotective agents. Based on an integrative model including 48,068 older adults, the main findings were: (1) biomarkers reflecting the five domains of IC can predict biological age with an acceptable error of 5.3 years, whereas among the biomarkers, vitality (i.e., handgrip strength) showed the greatest importance, followed by cognitive and sensory domains. (2) The most important geroprotective factor was the frequency of engaging in physical activity, which plays a significant role, particularly among females with comorbidities who do not engage in vigorous physical activity; and (3) smoking habits negatively impacted IC-age. These results support our hypothesis that IC-age differs between healthy individuals and those with multimorbidity, with geroprotective agents influencing this differentiation. They provide a comprehensive perspective on the lifespan trajectory of IC-age and highlight the critical role of lifestyle and geroprotective factors in shaping it.

Impact of multimorbidity and geroprotective factors on IC-age

The IC-age was significantly influenced by the presence of morbidity. Healthy individuals exhibited a lower IC-age compared to those with one or more comorbidities across all age groups up to 90 years old, beyond which the differences diminished. Regarding the individuals with a single disease, Alzheimer's disease, followed by hip fracture and Parkinson's disease, had the greatest impact on IC-age (see [online supplementary material Figure S3](#)). Previous studies have shown that IC is influenced by various factors, including demographic characteristics, socioeconomic status, lifestyle habits, and disease conditions (Beyene et al., 2024; Wei et al., 2024). Key factors associated with IC older adults include sex, marital status, occupational status, education, income/wealth, chronic diseases, smoking, alcohol consumption, and physical activity (Wei et al., 2024). In our study, the frequency of both moderate and vigorous physical activity emerged as the most influential geroprotective factor. Notably, our data indicated that engaging in vigorous physical activity, even as infrequently as once to three times per month, was better than not engaging at all, highlighting its potential benefits for healthy aging. This underscores the importance of promoting vigorous physical activity, even at low frequencies, as a valuable intervention for healthy aging. Previous studies have reported that higher levels of physical activity are associated with better IC, and that it tends to decline more rapidly in inactive individuals compared to those who are physically active (Raffin et al., 2025). Our results align with the existing literature and emphasize the particular importance of the frequency of vigorous physical activity in preserving IC-age (Beyene et al., 2024; Wei et al., 2024). Physical exercise may promote healthy aging by maintaining insulin sensitivity and functional skeletal muscle mass, reducing fat mass, and alleviating adipose tissue inflammation (Cartee et al., 2016). Remarkably, vigorous physical activity has been positively associated with cardiorespiratory fitness, while

moderate physical activity has shown beneficial associations with body fat (Drenowatz et al., 2016). A previous study from the UK Biobank cohort reported that 15–20 min of vigorous activity per week, even when accumulated in short bouts, was associated with reduced mortality and lower incidence of cardiovascular disease and cancer (Ahmadi et al., 2022). These findings hold significant public health and clinical relevance, mainly since lack of time is the most common barrier to regular physical activity, including age, sex, ethnicity, and health conditions (Cerin et al., 2010). Therefore, moderate and vigorous

activities should be endorsed in clinical and public health activity guidelines to maximize the population benefits of physical activity on IC-age (Gebel et al., 2015). Based on our results, the vitality and cognition domains were the most influential in estimating IC-age. Hence, future studies should investigate whether different strength training programs, alone or combined with cognitive interventions (Zhu et al., 2016), may help reverse or reduce IC-age.

Dietary habits have been identified as a protective factor for IC in older adults. Although our study found a trend toward significance regarding the number of diseases (comorbidities), it did not reach statistical significance. Dietary habits are also related to obesity, which is negatively associated with IC (Cesari et al., 2022; Lin et al., 2023; Yeung et al., 2022). Similarly, sleep problems have been reported to affect healthy aging negatively (Chang et al., 2023), but in our study, they did not reach statistical significance. However, the nature of the dataset limited our ability to explore dietary and sleep habits in depth. As both factors require more detailed assessment through specific instruments, these results should be interpreted with caution.

Regarding the negative lifestyle behaviors, in our study, smoking showed a significant negative impact on IC. Smoking is recognized as an aging accelerator that reduces life span and disease-free life (Bernhard et al., 2007). Our results are aligned with previous studies that have also reported the negative impact of smoking on IC (Muneera et al., 2022). In general, smokers (including asymptomatic ones) exhibit weaker and less fatigue-resistant muscles compared to nonsmokers, with cigarette smoke directly contributing to skeletal muscle dysfunction (i.e., decreased muscle mass and endurance) (Degens et al., 2015). These effects are further aggravated by the physical inactivity commonly observed among smokers. Smoking habits should be considered a critical negative lifestyle behavior that impacts the IC-age, independent of the number of diseases.

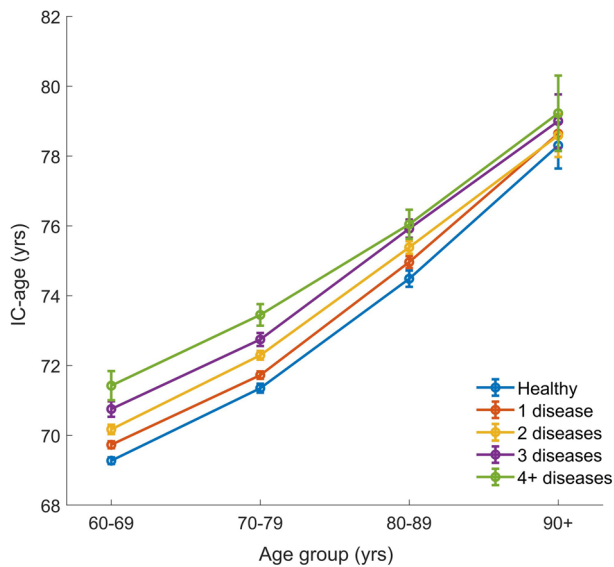


Figure 2. Intrinsic capacity age across age groups stratified by the number of diseases. Note. Lines represent group means, and error bars indicate the 95% CI. Groups include healthy individuals and those with 1, 2, 3, or 4+ diseases. IC-age = intrinsic capacity age.

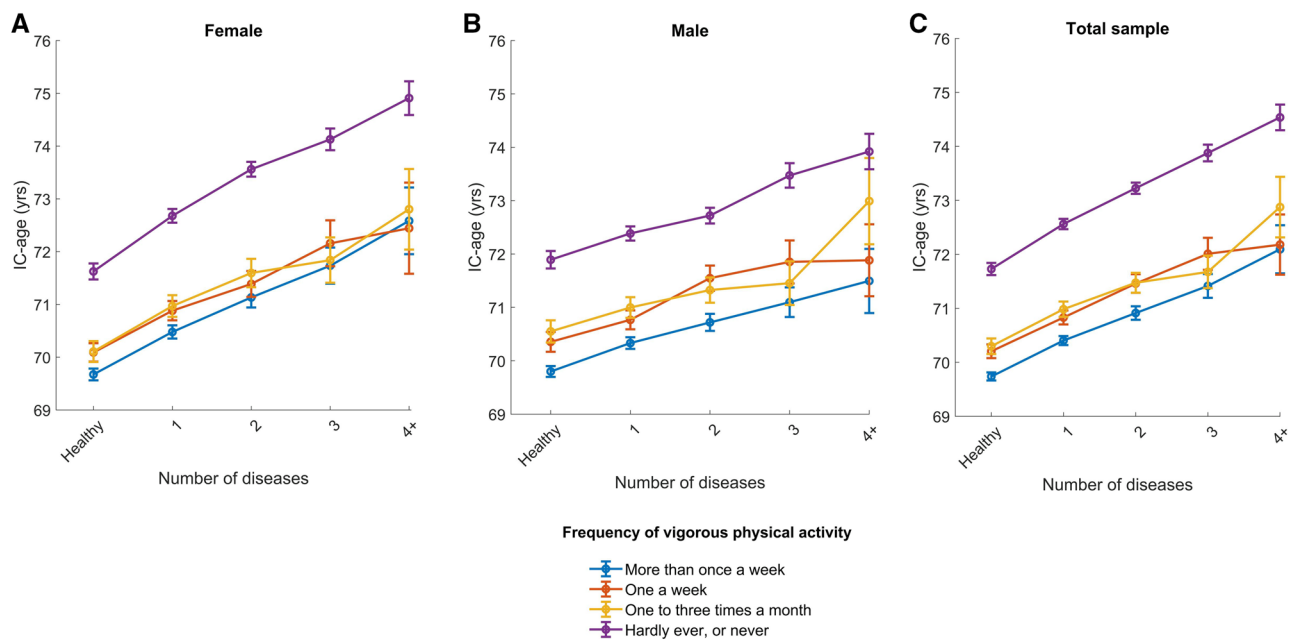


Figure 3. Intrinsic capacity age stratified by the number of diseases and frequency of vigorous physical activity. (A) Females, (B) males, and the (C) total sample.

Note. Lines represent group means, and error bars indicate the 95% CI. Vigorous physical activity frequency is categorized as more than once a week, once a week, one to three times a month, and hardly ever or never. IC-age = intrinsic capacity age.

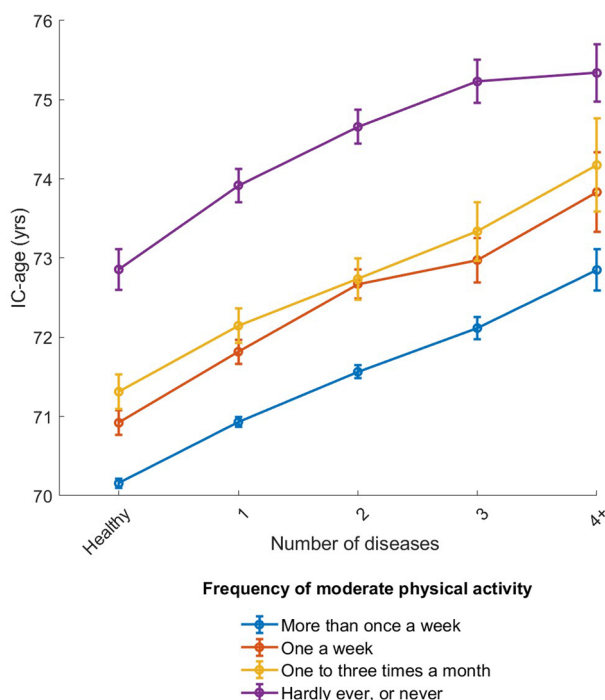


Figure 4. Intrinsic capacity age stratified by the number of diseases and frequency of moderate physical activity.
Note. Lines represent group means, and error bars indicate the 95% CI. Moderate physical activity frequency is categorized as more than once a week, once a week, one to three times a month, and hardly ever or never. IC-age = intrinsic capacity age.

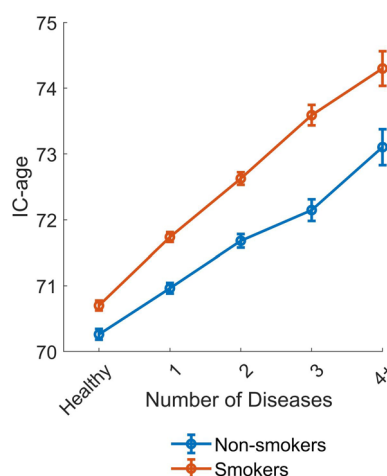


Figure 5. Intrinsic capacity age stratified by the number of diseases and smoking status.
Note. Lines represent group means, and error bars indicate the 95% CI. IC-age = intrinsic capacity age.

Another relevant point is the interaction between females and males with the number of diseases and the frequency of engaging in vigorous physical activity. Females showed a more significant impact from the absence of vigorous physical activity. This difference may be explained by previous studies showing that women generally have lower IC than men, particularly in vitality, psychological, and locomotor domains (Aliberti et al., 2022; M. Chen et al., 2025; Huang et al., 2024). These differences are thought to be attributed to physiological factors (e.g., muscle mass, strength, and endurance influenced by hormonal factors) and social contexts (e.g., socioeconomic status

and caregiving responsibilities for family members) that impact functionality and healthy aging (Aliberti et al., 2022; Huang et al., 2024). More studies are needed to understand the mechanisms that involve differences between males and females in IC.

Clinical and public health value of IC-age

Longitudinal studies have associated IC with frailty, social frailty, chronic diseases, fracture, and falls (Pan et al., 2024; Zhao et al., 2024), proposed its potential predictive value for adverse health outcomes (Zhao et al., 2024). The integrative model developed in this study represents a novel approach to predicting IC-age based on five domains of IC. The observed associations between IC-age, multimorbidity, physical activity, and lifestyle behaviors underscore its potential as a robust tool for understanding and monitoring intrinsic capacity in older adults. This model offers a new perspective for geroscience and public health by leveraging accessible biomarkers such as hand-grip strength, cognitive performance, and sensory health, making it suitable for routine clinical assessments without the need for invasive procedures or high-cost technologies.

Although achieving a prediction error of 5.3 years of IC-age, this accuracy is comparable to other biological age estimation models using genetic biomarkers, such as telomere length, which have reported prediction errors ranging from 3 to 6 years (R. Chen et al., 2023; Galkin et al., 2020; Mamoshina et al., 2018). Moreover, this level of accuracy aligns with findings from broader biological age prediction studies, which report MAE values ranging between 3.5 and 7.7 years across different biomarkers, including brain age, retinal age, and blood-based biomarkers (R. Chen et al., 2023). Given the potential public health relevance of IC-age, future research could incorporate epigenetic data (e.g., telomere length) alongside IC domains to improve the accuracy (Diebel & Rockwood, 2021; Galkin et al., 2020; Moqri et al., 2024). However, it is important to note that telomere length decreases with chronological age in a non-linear manner (Ye et al., 2023). For example, based on systematic review and meta-analysis the correlation between telomere length and age has been reported to be stronger in children ($r = -0.41$) than in adults ($r = -0.15$), suggesting that chronological age becomes a less dominant factor in explaining telomere attrition as individuals grow older (Ye et al., 2023).

The main strength of our study is the large dataset, comprising 48,068 participants from 27 European countries, which ensures robust and representative findings. Including biomarkers from all five intrinsic capacity domains provides a comprehensive and holistic perspective on aging. Furthermore, the analysis included validation of the model by testing interactions among sex, comorbidities, geroprotectors agents, and lifestyle factors, offering nuanced insights into the trajectories of IC-age. Despite the consistent group differences and interaction effects observed—supporting the validity of the IC-age model—longitudinal studies are still necessary to confirm its clinical applicability and public health relevance, particularly in relation to the meaning and implications of the prediction error. Additionally, longitudinal designs should be employed to assess how changes in IC-age correspond to shifts in both biological and chronological age.

In summary, IC-age represents a novel, cost-effective, and clinically applicable relevant functional biomarker of aging, with promising implications for early risk stratification, personalized interventions, and the development of geroprotective strategies tailored to aging populations.

Limitations

Our study has several limitations to be considered when interpreting the findings. First, cross-sectional design limits the ability to establish causal relationships between IC, geroprotective agents, and aging outcomes. Second, the nature of self-reported data introduces potential bias, particularly in chronic diseases and lifestyle habits. Such data are susceptible to recall and reporting inaccuracies, potentially affecting the IC-age and its associations. Additionally, the lack of detailed measures for dietary habits and sleep quality restricts exploring their impact on IC-age. Third, the variables selected to estimate IC-age were based on their availability within the SHARE dataset; therefore, the uneven distribution may not be fully representative of all domains. For example, the locomotion domain was limited to a single indicator—difficulty navigating within a room—and only a small number of individuals in the sample reported this condition. Other studies have utilized walking speed and chair stand test domain of locomotion (George et al., 2021). As a result, this aspect was not fully represented in our study and should be considered when interpreting the results. Lastly, conclusions for individuals over 90 years old are less robust due to smaller sample sizes and reduced differences observed in this age group.

Conclusion

This model demonstrates that IC domains can estimate biological age and distinguish individuals based on their comorbidities. It also underscores the role of physical activity as a key geroprotective factor, with vigorous physical activity in females with comorbidities showing the most pronounced protective effect on IC-age. These results validate the concept of IC-age as a comprehensive measure of aging and highlight its potential to inform personalized interventions and public health strategies. Future research should focus on external validation of IC-age in non-European populations and assessment of its predictive value for clinical and public health outcomes through longitudinal follow-up studies.

Supplementary material

Supplementary data are available at *The Gerontologist* online.

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Conflict of interest

None declared.

Data availability

This study uses data from the *Survey of Health, Ageing and Retirement in Europe (SHARE)* project (10.1093/ije/dyt088). SHARE data are available to registered researchers via <https://share-eric.eu/data>. Analytic code is available upon reasonable request. The study was not pre-registered.

Authors contribution

Carlos Cruz-Montecinos and Rodrigo Núñez-Cortés contributed to the conception and design of the study. Carlos Cruz-Montecinos conducted the model, and Carlos

Cruz-Montecinos and Rodrigo Núñez-Cortés performed the data analysis. Carlos Cruz-Montecinos and Rodrigo Núñez-Cortés drafted the manuscript. Carlos Cruz-Montecinos, Joaquín Calatayud, Lars Louis Andersen, Rubén López-Bueno, Rodrigo Torres-Castro, Luis Peñailillo, Fernando Diefenthaler, and Rodrigo Núñez-Cortés made substantial contributions to the interpretation of data, critically revised the manuscript, and approved the final version for publication.

Ethics approval and consent to participate

Ethical approval for the collection of the data used in this study and consent to participate was obtained by the Survey of Health, Ageing and Retirement in Europe [SHARE] Project (10.1093/IJE/DYT088).

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